

FREEDOM INTERNATIONAL SCHOOL

PRACTICE PAPER - 1

MATHEMATICS

CLASS: XI

MAX MARKS: 80

TIME: 3 HOURS

GENERAL INSTRUCTIONS:

1. This question paper contains - **five sections A, B, C, D and E**. Each section is compulsory. However, there are internal choices in some questions.
2. **Section-A** has **18 MCQs** and **02 Assertion-Reason** based questions of **1 mark** each.
3. **Section-B** has **5 Very Short Answer (VSA)** type questions of **2 marks** each.
4. **Section-C** has **6 Short Answer (SA)** type questions of **3 marks** each.
5. **Section-D** has **4 Long Answer (LA)** type questions of **5 marks** each.
6. **Section-E** has **3 source based/case based/passage based/integrated units of assessment** (4 marks each) with sub parts.
7. Internal choice is provided in **2 questions in Section-B, 2 questions in Section-C, 2 questions in Section-D**. You must attempt only one alternative in all such questions.

SECTION – A

(All questions are compulsory. No internal choice is provided in this section)

Each question carries 1 mark.

1. If $\tan \alpha = \frac{x}{x+1}$ and $\tan \beta = \frac{1}{2x+1}$ then $\alpha + \beta$ is equal to
a) $\frac{\pi}{2}$ b) $\frac{\pi}{3}$ c) $\frac{\pi}{4}$ d) $\frac{\pi}{6}$
2. Two cards are drawn successively without replacement from a well-shuffled pack of 52 cards. The probability of drawing two aces is
a) $\frac{1}{221}$ b) $\frac{1}{26}$ c) $\frac{1}{13}$ d) $\frac{4}{223}$
3. In a group of 40 students, 26 take tea, 18 take coffee and 8 take neither of the two. How many take both tea and coffee?
a) 10 b) 12 c) 14 d) 16
4. If $N = 50$, $\sum x_i^2 = 2400$ and $\sum x_i = 250$, then, $\sigma^2 =$
a) 20 b) 21 c) 22 d) 23
5. The reflection of the point (α, β, γ) in the $xy -$ plane is
a) $(\alpha, \beta, -\gamma)$ b) $(0, 0, \gamma)$ c) $(\alpha, \beta, 0)$ d) $(-\alpha, -\beta, -\gamma)$
6. $(\sqrt{5} + 1)^4 + (\sqrt{5} - 1)^4$ is _____.
a) an irrational number b) a negative real number
c) a rational number d) a negative integer
7. A box contains 10 good articles and 6 with defects. One item is drawn at random. The probability that it is either good or has a defect is
a) $\frac{64}{64}$ b) $\frac{49}{64}$ c) $\frac{40}{64}$ d) $\frac{24}{64}$
8. The solution set for $|3x - 2| \leq \frac{1}{2}$ is
a) $\left[\frac{2}{3}, \frac{5}{6}\right]$ b) $\left[-\frac{2}{3}, \frac{2}{3}\right]$ c) $\left[\frac{1}{2}, \frac{5}{6}\right]$ d) $\left[-\frac{1}{2}, \frac{5}{6}\right]$

9. If R is a relation from $\{11, 12, 13\}$ to $\{8, 10, 12\}$ defined by $y = x - 3$, then R^{-1} is
 a) $\{(10,13), (12,10)\}$ b) $\{(10,13), (8,11), (12,10)\}$
 c) $\{(11,8), (13,10)\}$ d) $\{(8,11), (10,13)\}$
10. If $8 \sin \frac{x}{8} \cos \frac{x}{2} \cos \frac{x}{4} \cos \frac{x}{8}$ is equal to
 a) $8 \cos x$ b) $8 \sin x$ c) $\cos x$ d) $\sin x$
11. In how many ways can 4 women draw water from 4 taps if no tap remains unused?
 a) 18 b) 120 c) 22 d) 24
12. If $y = \frac{1 + \frac{1}{x^2}}{1 - \frac{1}{x^2}}$, find $\frac{dy}{dx}$.
 a) $\frac{-4x}{x^2-1}$ b) $\frac{1-x^2}{4x}$ c) $\frac{-4x}{(x^2-1)^2}$ d) $\frac{4x}{x^2-1}$
13. In a circle of radius 14 cm, an arc subtends an angle of 36° at the centre. The length of the arc is
 a) 7.7 cm b) 8.8 cm c) 9.1 cm d) 6.6 cm
14. If a, b, c are in A.P. and x, y, z are in G.P. then the value of $x^{b-c} \cdot y^{c-a} \cdot z^{a-b}$ is
 a) 0 b) 1 c) xyz d) $x^a y^b z^c$
15. If the circle $x^2 + y^2 + 2ax + 8y + 16 = 0$ touches the x -axis, then the value of a is
 a) ± 16 b) ± 4 c) ± 8 d) ± 1
16. $\lim_{x \rightarrow 0} \frac{\cos ax - \cos bx}{\cos cx - \cos dx}$
 a) $\frac{a^2-b^2}{c^2-d^2}$ b) $\frac{b^2-a^2}{c^2-d^2}$ c) $\frac{a^2-b^2}{d^2-c^2}$ d) $\frac{a^2+b^2}{c^2+d^2}$
17. If $\cot \theta = \frac{1}{2}$ and $\sec \phi = \frac{-5}{3}$, where θ lies in III quadrant and ϕ in II quadrant, then $\tan(\theta + \phi) = ?$
 a) $\frac{-6}{11}$ b) $\frac{5}{11}$ c) $\frac{2}{11}$ d) $\frac{10}{11}$
18. The complex number z which satisfies the condition $\left| \frac{i+z}{i-z} \right| = 1$ lies on
 a) circle $x^2 + y^2 = 1$ b) the x -axis
 c) the y -axis d) the line $x + y = 1$

ASSERTION-REASON BASED QUESTIONS

In the following questions, a statement of Assertion (A) is followed by a statement of reason (R). Choose the correct answer out of the following choices.

- a) Both A and R are true and R is the correct explanation of A.
 b) Both A and R are true but R is not the correct explanation of A.
 c) A is true but R is false.
 d) A is false but R is true.

19. **Assertion (A):** The sum of series $(\sqrt{2} + 1), 1, (\sqrt{2} - 1), (\sqrt{2} - 1)^2 \dots$ upto ∞ , is $4 + 3\sqrt{2}$.

Reason (R): For an infinite geometric progression $S = \frac{a}{1-r}$; $|r| < 1$

20. Assertion (A): $y + 2 = 0$ and $\sqrt{3}x - y - 3\sqrt{3} - 2 = 0$ are the equations of the lines passing through the point $(3, -2)$ and inclined at an angle of 60° to the line $\sqrt{3}x + y = 1$.

Reason (R): The equation of a straight line through the point $A(x_1, y_1)$ and having slope m is $y - y_1 = \frac{1}{m}(x - x_1)$.

SECTION – B

(All questions are compulsory. In case of internal choice, attempt any one question only)

Each question carries 2 marks.

21. Find the roots of the equation $3x^2 + 3xi + 10 = 0$.

22. If $A = \{1, 2, 3\}$, $B = \{4\}$ and $C = \{5\}$, then verify that

$$A \times (B - C) = (A \times B) - (A \times C).$$

23. Evaluate $\lim_{x \rightarrow 1} \frac{x^2 - \sqrt{x}}{\sqrt{x} - 1}$.

24. Find the domain and range of the function, $f(x) = \frac{x^2 - 16}{x - 4}$.

OR

Find the domain and range of the function $f(x) = \sqrt{x - [x]}$

25. Find the equation of the hyperbola having foci $(\pm 3\sqrt{5}, 0)$. The latus rectum is of length 8 units.

OR

Find the lengths of the major axis, coordinates of the foci, and eccentricity of the ellipse : $3x^2 + 2y^2 = 6$.

SECTION – C

(All questions are compulsory. In case of internal choice, attempt any one question only)

Each question carries 3 marks.

26. If $(a + ib) = \frac{c+i}{c-i}$, where c is real, prove that $a^2 + b^2 = 1$ and $\frac{b}{a} = \frac{2c}{c^2 - 1}$.

27. Find a , if the coefficient of x^2 and x^3 in the expansion $(3 + ax)^9$ are equal.

OR

Find the fourth term from the end in the expansion $\left(\frac{3}{x^2} - \frac{x^3}{6}\right)^7$.

28. a) Draw the graph of the Greatest Integer Function.

b) Write domain and range of the Greatest Integer Function.

29. Find the point in yz – plane which is equidistant from the points $A(3, 2, -1)$, $B(1, -1, 0)$ and $C(2, 1, 2)$.

30. a) Using product rule of derivatives, find the derivative of the function given by $f(x) = (x \operatorname{cosec} x - \cot 2x)(x^3 \log x - e^{4x})$.

OR

b) Using quotient rule of derivatives, find the derivative of the function given by

$$f(x) = \frac{(\sec x - x \tan x)}{(x \tan x - \operatorname{cosec} x)}.$$

- 31 a) In how many ways can 5 girls and 3 boys be seated in a row with 11 chairs so that no two boys sit together?
- b) What is the number of ways in which a set of 5 cards be chosen out of a deck of 52 cards, if each set of 5 cards have exactly one ace?

SECTION – D

(All questions are compulsory. In case of internal choice, attempt any one question)

Each question carries 5 marks.

- 32 Find mean deviation about the mean for the data:

Height (cm)	95-105	105-115	115-125	125-135	135-145	145-155
No. of boys	9	16	23	30	12	10

- 33 Prove that: $\cos^3 x \sin 3x + \sin^3 x \cos 3x = \frac{3}{4} \sin 4x$.

OR

Prove that:

$$\cos \alpha + \cos \beta + \cos \gamma + \cos(\alpha + \beta + \gamma) = 4 \cos\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\beta + \gamma}{2}\right) \cos\left(\frac{\alpha + \gamma}{2}\right).$$

- 34 A) If p_1 and p_2 are the lengths of perpendiculars from the origin to the line $x \sec \theta + y \csc \theta = a$ and $x \cos \theta - y \sin \theta = a \cos 2\theta$ respectively then prove that $4p_1^2 + p_2^2 = a^2$

OR

- B) a) Find the equation of the line parallel to the line $2x - y - 9 = 0$ and passing through the point of intersection of the lines $5x + y + 4 = 0$ and $2x + 3y = 1$.
- b) Reduce the equation $3x - 2y + 6 = 0$ to the intercept form and find the x and y intercepts.

- 35 Differentiate $\sqrt{\cot 3x}$ using the first principle method.

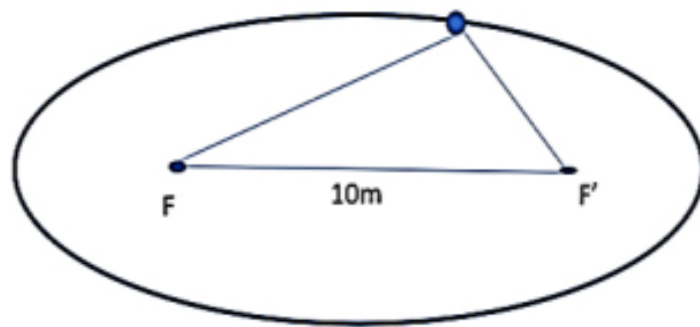
SECTION – E

(All questions are compulsory. In case of internal choice, attempt any one question only)

Each question carries 4 marks.

- 36 Case Study -1: A farmer wishes to install 2 handpumps in his field for watering.





A farmer moves in the field while watering in such a way that the sum of distances between the farmer and each handpump is always 26 m. Also, the distance between the hand pumps is 10 m.

Based on the above information, answer the following questions:

- a) Name the curve traced by the farmer and hence find the foci of the curve. [1]
- b) Find the equation of the curve traced by the farmer. [1]
- c) i) Find the length of the major axis, minor axis and eccentricity of the curve formed. [2]

OR

- ii) Find the length of the latus rectum and the distance between the foci. [2]

37 Case Study -2: Two students Ankit and Vinod appeared in an examination. The probability that Ankit will qualify for the examination is 0.05 and Vinod will qualify is 0.10. The probability that both will qualify is 0.02.



Based on the above information, answer the following questions:

- a) Find the probability that at least one of them will qualify for the exam. [1]
- b) Find the probability that at least one of them will not qualify for the exam. [1]
- c) i) Find the probability that both Ankit and Vinod will not qualify the exam. [2]

OR

- ii) Find the probability that only one of them will qualify the exam. [2]

38 Case Study -3: In a city school during the admission to class XI, 18 students took English, 23 students took Hindi and 24 students took Sanskrit. Of these, 13 took both Hindi and Sanskrit, 12 took both English and Hindi and 11 took both English and Sanskrit. Due to the request made by some students, the school authorities decided that 6 students will be offered all three languages.



- a) i) How many students took Sanskrit but not Hindi? [1]
ii) How many students took exactly one of the three subjects? [1]
- b) i) How many students took exactly two of the three subjects? [1]
ii) How many students took Hindi but not Sanskrit? [1]

*****BEST OF LUCK*****